

Assignment: Diagnostic Analysis using Python

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1. Background/context of the business scenario

The National Health Service (NHS) is facing increasing pressure to manage rising demand while ensuring efficient allocation of staff, infrastructure and financial resources. Stakeholders are divided on whether capacity should be expanded or whether existing resources can be better optimised. Additionally, missed appointments represent a significant financial and social cost, disproportionately affecting vulnerable populations. This project aims to explore internal NHS appointment data and external Twitter data to assess service utilisation trends, staff capacity and appointment attendance patterns. The goal is to provide data driven insights to support budgeting, workforce planning, and service optimisation decisions.

2. Analytical approach

Prior to analysis approach was planned, the main aim of the analysis is to plan budgets and expand infrastructure effectively with sufficient justification for stakeholders that feel investment is currently sufficient. Missing appointments costs the NHS significantly reducing missed appointments would be hugely beneficial for the NHS.

Data Import and Exploration

The first step was cleaning, viewing, exploring and analysing the three datasets - actual_duration.csv ,appointments_regional.csv and national_categories.xlsx. Later in the report twitter sample data was also analysed.

Analysis was conducted in Python using Pandas and NumPy for data manipulation and Seaborn/Matplotlib for visualisation in Jupyter Notebook

Initial exploratory analysis included reviewing column structures, checking missing values using `.isnull().sum()`, assessing unique categories using `.nunique()` and identifying date ranges using `.min()` and `.max()`

Date variables were converted to datetime format where necessary. The dataset was filtered from August 2021 onwards to ensure consistent and recent analysis.

Data Cleaning and Preparation

Categorical and numerical fields were formatted, data aggregated by month and derived variables such as daily utilisation were created.

Capacity and Utilisation Assessment

Monthly appointment totals were aggregated and divided by 30 to estimate average daily utilisation. Daily utilisation increased steadily from approximately 795,000 appointments per day in August 2021 to just over 1 million per day in October and November, before declining to around 838,000 in December. When compared to the

NHS planning benchmark of 1.2 million daily appointments, utilisation remained below maximum capacity but approached roughly 84-85% of the benchmark during peak autumn months. This indicates that while the system experienced heightened seasonal demand and operational pressure in October and November, it did not exceed planned capacity thresholds. The decline in December reflects expected seasonal variation, likely linked to reduced service activity during the holiday period.

	appointment_month	count_of_appointments	utilisation
0	2021-08-01	23852171	795072.4
1	2021-09-01	28522501	950750.0
2	2021-10-01	30303834	1010127.8
3	2021-11-01	30405070	1013502.3
4	2021-12-01	25140776	838025.9

Twitter Data Processing

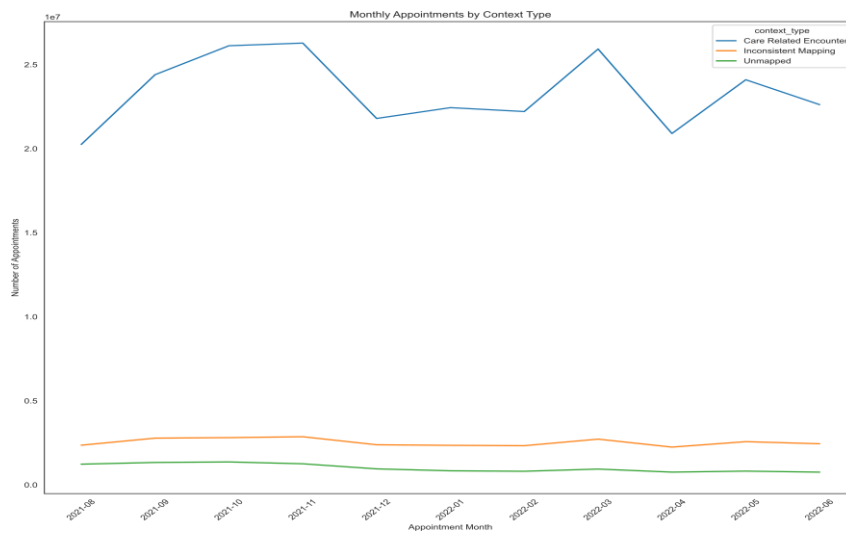
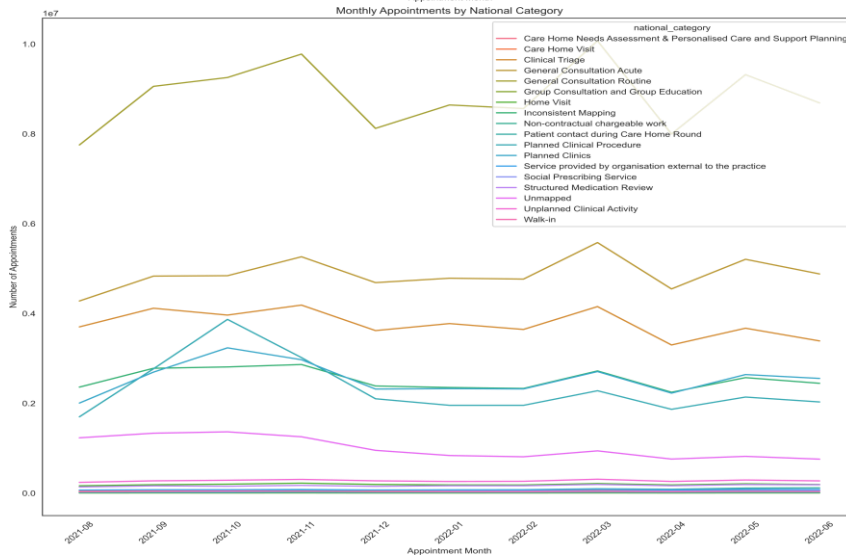
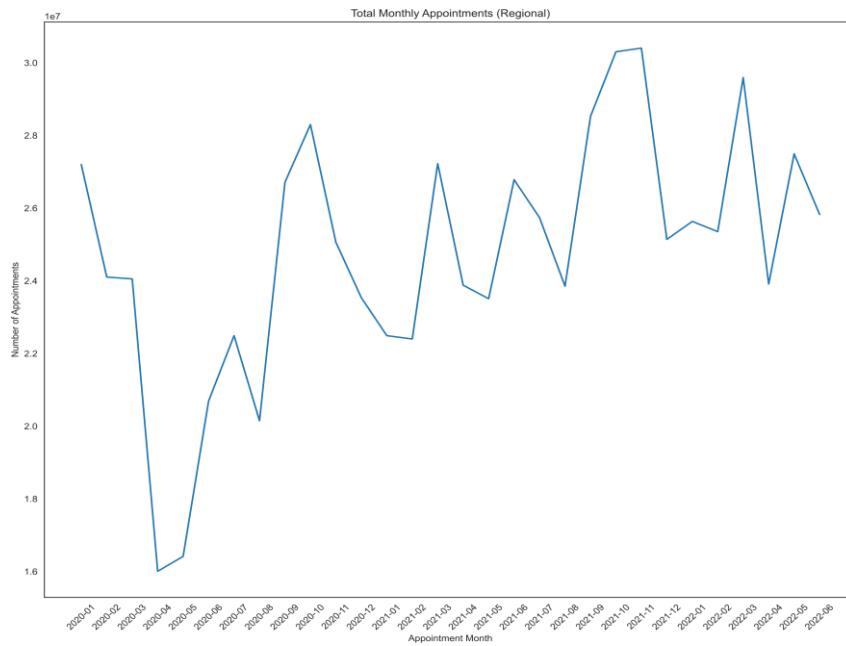
Text parsing methods were used to extract hashtags from tweet text fields. These were aggregated into a frequency data frame and visualised to identify trending healthcare concerns. This structured workflow ensured reproducibility and alignment between technical analysis and business objectives.

3. Visualisation and insights

Visualisations were selected to communicate trends clearly to both technical and business stakeholders.

Monthly Appointment Trends

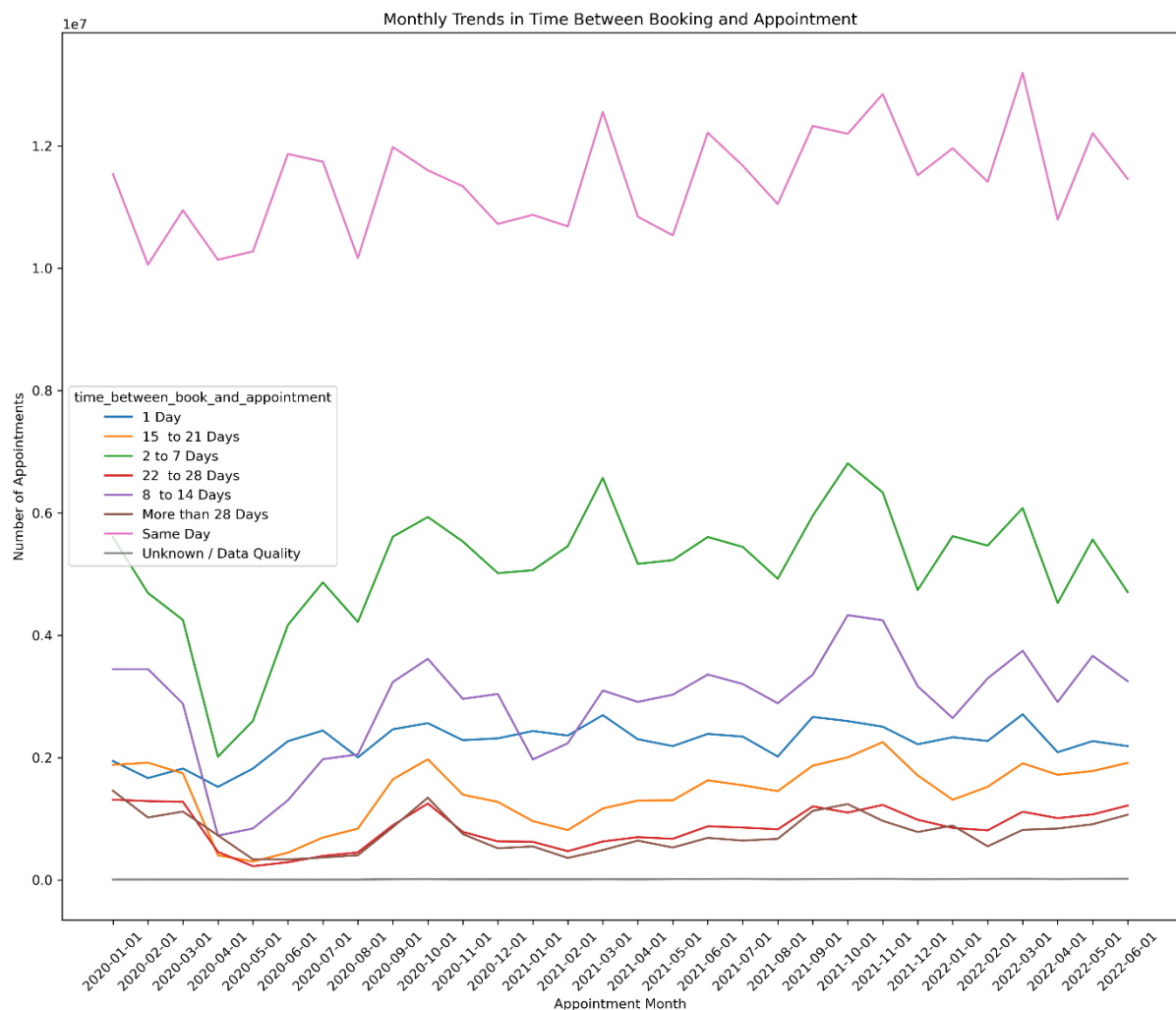
Line plots were used to visualise total monthly appointments. These revealed clear seasonal patterns, with peaks during October and November and lower volumes during March and April. This seasonal fluctuation is critical for workforce and budgeting decisions. A general consultation routine is the most popular national category and the most common context type is care related encounters. Line plots can be seen below -



Time Between Booking and Appointment

There is a clear upward trend in appointment volumes following the COVID disruption in early 2020. Same day appointments consistently represent the largest proportion of activity, indicating a system heavily oriented toward rapid access and urgent care delivery. Short wait appointments (2–7 days) also demonstrate sustained growth throughout 2021, reflecting recovery of routine services. However, modest increases in longer waiting categories during peak months suggest periods of operational strain and possible backlog accumulation. Overall, the system shows resilience and recovery but evidence of seasonal pressure during high demand periods.

Attended appointments represented the majority, while DNAs remained a stable minority, representing avoidable cost. Targeted interventions such as reminders and flexible rebooking may be more effective than punitive measures.



Missed appointments

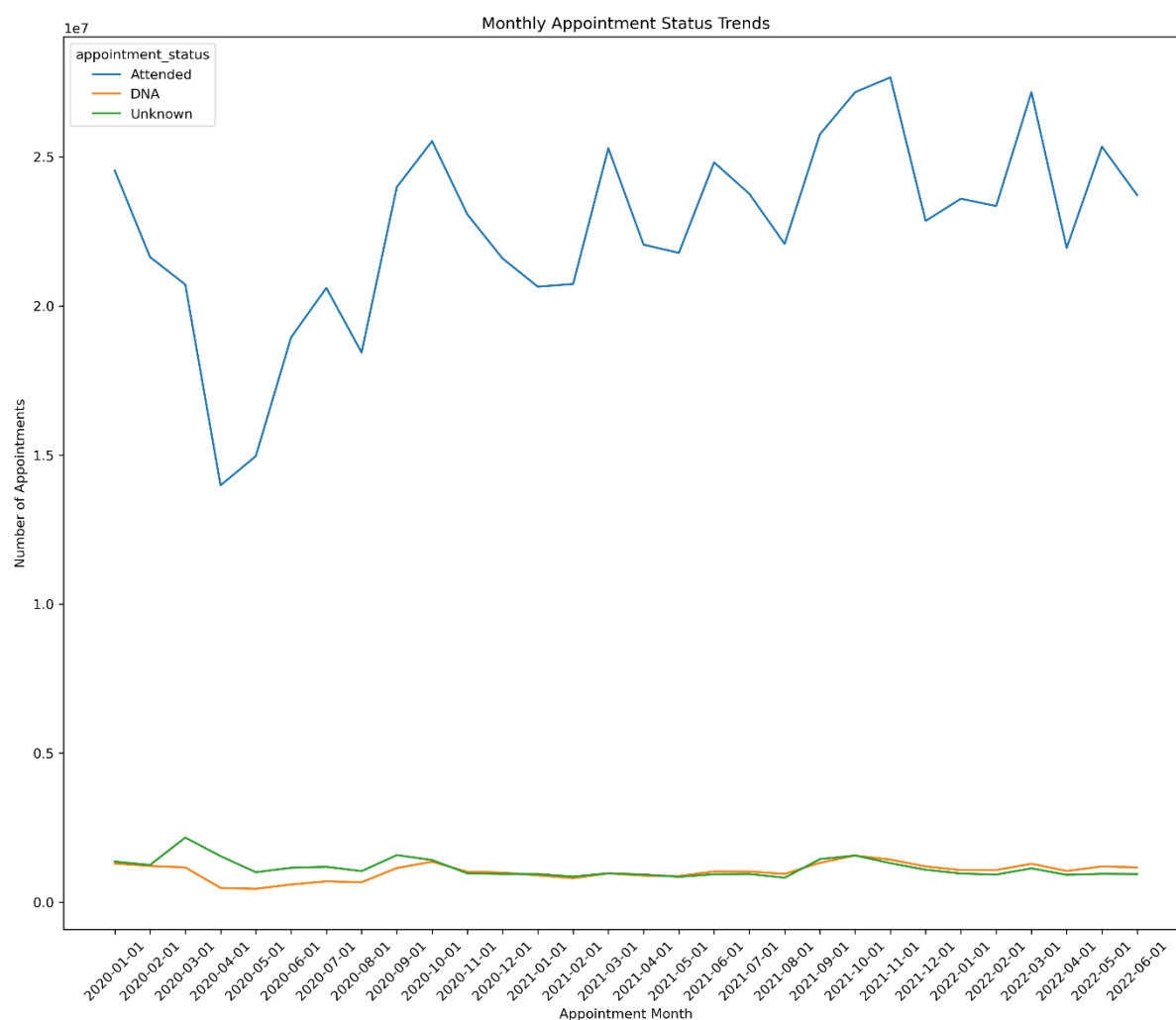
Most missed appointments occur in GP services, which is expected given that GP appointments are the most frequent.

Attended appointments consistently account for the vast majority of activity across the entire period. Although there is a sharp decline around April 2020 likely reflecting the initial COVID disruption volumes recover steadily and peak throughout late 2021 and early 2022. This suggests strong system resilience and restoration of service capacity following pandemic restrictions.

DNA (Did Not Attend) appointments remain significantly lower than attended appointments but show relatively stable volumes over time. There is no evidence of a sustained upward trend in DNAs, indicating that non attendance did not worsen structurally despite fluctuations in overall demand. However, given the scale of total monthly appointments, even this small proportion represents substantial lost clinical capacity.

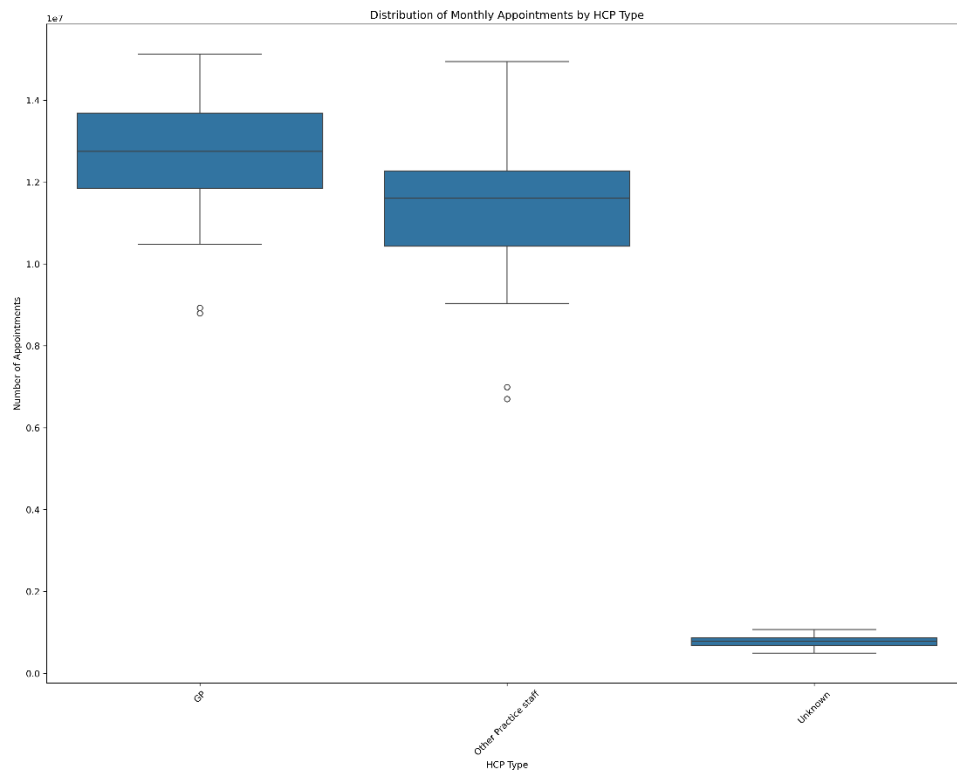
The “Unknown” category follows a similar low volume and stable pattern, suggesting consistent data recording practices. However, improvement in data collection to avoid this category would make analysis more accurate.

Overall, the data indicates stable attendance behaviour over time, with variations largely driven by overall service demand rather than changes in patient attendance patterns.



Healthcare Professional (HCP) Trends

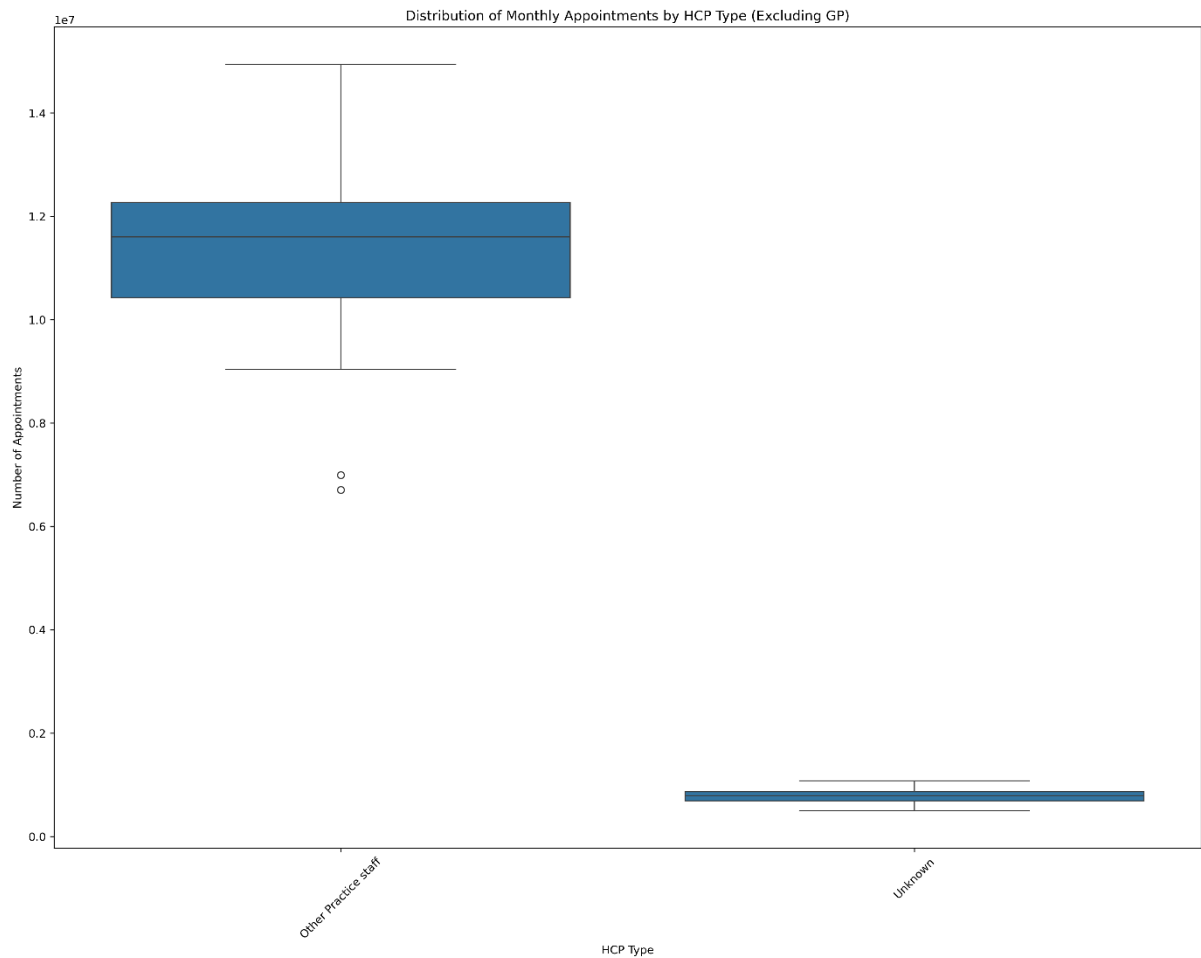
GPs account for the highest volume of monthly appointments, with consistently greater median activity than all other HCP types. Other practice staff contribute a substantial but lower share, indicating partial workload distribution across the primary care workforce. The “Unknown” category remains minimal and stable. Overall, the distribution shows that primary care delivery remains predominantly GP led, despite evidence of some role diversification. This suggests potential redistribution opportunities to relieve GP pressure.



Service Settings Comparison

Boxplots revealed that GP settings show higher volume and variability, excluding GP data clarifies patterns among smaller service settings, other practice staff handle the vast majority of monthly appointments, with consistently high volumes and some variability.

The “unknown” category contributes very few appointments and remains relatively stable. This shows that, outside of GPs, workload is overwhelmingly carried by other practice staff.



Twitter Insights

Hashtag analysis identified recurring themes such as #healthcare that was used 785 times and #health used 84 times, attempting to indicate public concern around access and waiting times through twitter data is limited. Other less used hashtags include #nurseproblems but greater understanding of these problems will require content consumption of all tweets. Social media data provides contextual sentiment that complements operational metrics but deeper analysis would be needed.

4. Patterns and predictions

Seasonal peaks in October–November require workforce planning to shift resources rather than increase investment.

There is a high dependency on GP services. Utilisation of non GP healthcare professionals where clinically appropriate to reduce demand on one service, increasing usage of the primary care network and extended access provision.

There are persistent but stable missed appointment levels, there is a high but not excessive overall capacity utilisation. An improvement in the appointment reminder system and targeted communication may reduce DNAs.

Unmapped appointments need to be categorised for more accurate analysis going forward as these account for sufficient data entries, 1054 in August 2021 alone.

Enhanced social media tracking could help identify recurring service pressures.

Further analysis could include predictive modelling of seasonal demand to understand drivers of missed appointments as current data is limited on specific reasons.

To conclude, prior to increasing investment the NHS should work to shift current resources, seasonal workforce planning to increase staffing during busier months of October and November and utilisation of non GP healthcare professional services where clinically appropriate should be implemented.

Investment, however, should be done in automated appointment reminders as it is possible long wait times are the reason behind missed appointments. Investment in greater social media analysis would also help identify recurring issues split by demographics such as age and location.